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Discrete-time Pole Placement with Stable Controller

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A method to compute a solution to a polynomial Diophantine equation provides a procedure to design stable discrete-time pole placement controllers, and to solve a simultaneous partial pole placement problem.

Key Words—Pole placement; discrete-time system; linear systems; digital control; interpolation; stability.

Abstract—The classical polynomial approach to pole placement involves the computation of a particular solution of a given Diophantine equation. The solution corresponds to a proper controller whose order is fixed by the number of nonminimum phase zeros of the plant transfer function, by its order, and by the desired closed-loop characteristic polynomial. This controller might be unstable for some desired closed-loop characteristic polynomials, even though the plant can be stabilized by a stable controller. Now, unstable controllers are known to deteriorate closed-loop system performances, compared with stable ones. Hence, we propose a procedure to build, for a specific class of plants, a stable pole placement controller when the classical pole placement controller is unstable. The properties of both stable and unstable controllers are compared via simulations. It appears that, for the considered plants, the stable controller improves at least one of the following properties, compared with its unstable counterpart: step response transient, gain margin, measurement noise rejection. The results obtained with our procedure are also compared with the results obtained via Middleton's approach, which consists in modifying the desired closed-loop poles so as to obtain a stable and minimum phase controller with the classical pole placement method. Finally, our pole placement design with stable controller is applied to solve a particular simultaneous partial pole placement problem.

1. INTRODUCTION

THE POLYNOMIAL APPROACH to discrete-time pole placement has been thoroughly studied in Åström and Wittenmark (1984) for instance. Although this approach is quite popular among control engineers, and although it is widely used in adaptive control (Åström and Wittenmark, 1980, 1989), it is well known that the classical pole placement procedure might yield an unstable controller, even though the plant can be stabilized by a stable controller (Middleton and Goodwin, 1990). Now, unstable poles and nonminimum phase zeros in the loop deteriorate the properties of the closed-loop sensitivity and complementary sensitivity functions. This has first been shown by Freudenberg and Looze (1985), for continuous-time systems, and their results have been recently translated to discrete-time plants in Middleton and Goodwin (1990), Mohtadi (1990) and Middleton (1991). Middleton and Goodwin (1990) and Middleton (1991) also give time domain arguments justifying why controllers with poles and/or zeros inside the unit circle (in the z^{-1}-plane) should be avoided.

Several routes are possible in order not to obtain an unstable controller via a pole placement design. One of them consists in choosing the desired closed-loop characteristic polynomial in such a way that the classical pole placement procedure yields a stable controller. This idea has been pursued in Middleton and Goodwin (1990), where somewhat heuristic guidelines are given to choose adequately the desired closed-loop poles. Controllers designed with this approach will be called SCACP for "Stable Controller based on Adequate Characteristic Polynomial".

Here, we investigate a second route which yields a so-called SCI, namely "Stable Controller based on an Interpolation procedure". The justification of this name will become obvious later. For continuous-time systems, Youla (Youla *et al.*, 1974) developed a method to build a stable pole placement controller which amounts to an interpolation problem similar to ours. However, he considered feedback systems including both a compensator C and a feedback sensor F to be designed (see Fig. 1a), while we

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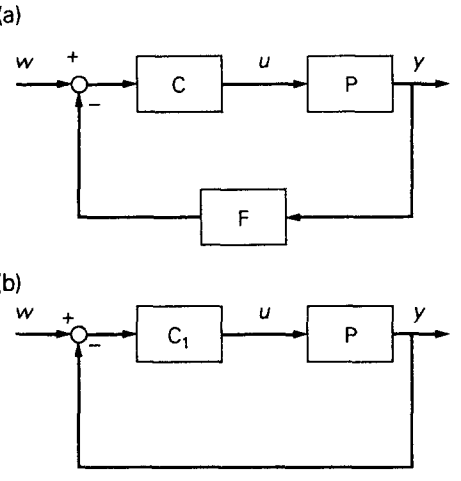


FIG. 1. (a) Feedback system with compensator and feedback sensor. (b) Classical feedback system with a one degree of freedom controller.

only consider a classical one degree of freedom controller C_1 (see Fig. 1b). Youla's procedure consists of two steps. He first designs, for the controlled system, a sensitivity function which can be achieved with a stable controller and a stable feedback sensor, without taking care of the poles of this function, save for the requirement that they are stable; then he uses this sensitivity function to design a stable controller and a stable feedback sensor that achieve a desired overall closed-loop transfer function. Our method also proceeds in two steps, but it uses a one degree of freedom controller as stated above. We first design a controller which places the closed-loop poles at a desired location. Then, if the resulting controller is unstable, we modify it in order to assure its stability, without moving the closed-loop poles. However, we allow the introduction of some closed-loop poles at infinity (in the z^{-1}-plane). Should we not allow these extra poles, we would have to use a two degree of freedom controller or a feedback sensor. Our approach is more in the lines of the classical literature on discrete-time pole placement, such as Åström and Wittenmark (1984) or Middleton and Goodwin (1991) because it is based on the resolution of a polynomial Diophantine equation.

A third route to design stable pole placement controllers could be developed on the basis of the work of Saeks *et al.* (1982), wherein the class of all the compensators achieving a desired closed-loop characteristic polynomial is parametrized in terms of an arbitrary polynomial whose degree is less than a fixed bound. It would consist in searching a stable controller in this class. One could even look for a stable controller which minimizes a given cost functional. However, the constraint on the stability of the controller is, to our knowledge, difficult to handle in a numerically efficient way. This approach will not be pursued here.

We shall compare the first two proposed design procedures with the classical pole placement approach, via simulations. This will illustrate the importance of using stable controllers.

Finally, we shall apply the design of a SCI to solve a particular simultaneous partial pole placement problem. This approach can be used to design a controller for a plant which can be switched into two operating modes, or for a nonlinear system which has been linearized at two equilibria. Emre (1983) attempted to stabilize a set of plants with a single controller in such a way that the closed-loop systems obtained for the different plants have the same fixed characteristic polynomial. He showed that this cannot always be achieved, even if the set only consists of two different plants. Moreover, when this problem has a solution, the closed-loop characteristic polynomial is entirely defined by the couple of plants; it cannot be chosen by the designer. Hence the need to look at partial pole placement to have some freedom in the choice of the closed-loop poles for simultaneous stabilization of SISO plants. Ghosh (1986) gives a method to place, with a single controller, part of the closed-loop poles at desired fixed locations and the remainder in a given region C_s (a self-conjugate open subset of the complex plane which intersects the real axis), and this, simultaneously for a couple of plants. The fixed closed-loop poles are not contained in the region C_s. One of the motivations for such a problem statement is the so-called dominant pole placement problem. On the other hand, Vidyasagar's results (1987b) give a procedure to design a controller that places the closed-loop poles of a couple of plants, in the regions C_{s1} for plant one and in C_{s2} for plant two. Here, C_{s1} and C_{s2} are stability regions of the same type as C_s. Moreover, $C_{s1} \subseteq C_{s2}$ and both C_{s1} and C_{s2} fulfil some additional technical conditions.

In this paper, we describe a solution to the following problem. Given two plants, find, if possible, a single controller that yields a closed-loop system with a fixed desired characteristic polynomial for the first plant, and a closed-loop system with poles in a desired region for the second plant. The differences with the work of Ghosh (1986) and Vidyasagar (1987b) will be explained.

This paper is organized as follows. In Section 2, we introduce some notations and we describe the class of plants. Pole placement design is reviewed in Section 3. The procedure to build a

stable controller is studied in Section 4 Simulation results are discussed in Section 5. Finally simultaneous partial pole placement is developed in Section 6.

2. PRELIMINARIES

2.1. *Notation*

q^{-1} is the backward shift operator,
λ is the transform variable associated with the operator q^{-1},
$\mathbb{P}_1$ is the closed unit disc in the complex plane,
R_∞ is the set of rational functions without poles in $\mathbb{P}_1$,
$\mathbb{P}$ is a closed simply connected domain which is the generalized region of stability. Here for the sake of simplicity $\mathbb{P}$ is a closed disc of radius $\alpha \geq 1$,
$\bar{\mathbb{P}}$ is the complement of $\mathbb{P}$ in $\mathbb{C}$,
$R_{\bar{\mathbb{P}}}$ is the set of rational functions with all their poles in $\bar{\mathbb{P}}$. A unit of $R_{\bar{\mathbb{P}}}$ is an element of $R_{\bar{\mathbb{P}}}$ whose inverse lies in $R_{\bar{\mathbb{P}}}$,

$$\|f(\lambda)\|_\infty = \sup_{\theta \in [0,2\pi]} |f(e^{j\theta})| \quad \text{where} \quad f(\lambda) \in R_\infty.$$

2.2. *The class of systems*

We consider the class of strictly proper discrete-time linear time invariant single-input–single-output (SISO) systems described by the following model:

$$A(q^{-1})y(t) = B(q^{-1})u(t), \tag{1}$$

where $A(q^{-1})$ and $B(q^{-1})$ are polynomials in the backward shift operator q^{-1} of degree na and nb, respectively, $B(0) = 0$ and without loss of generality we take $A(0) = 1$. $u(t)$ and $y(t)$ are, respectively the input and the output signals of the plant. We assume that $A(q^{-1})$ and $B(q^{-1})$ are coprime. Moreover, since we shall be dealing with pole placement by stable controllers in Sections 4 and 5, we only consider plants that can be stabilized by a stable controller in that part of the paper. Such plants are said to be strongly stabilizable.

3. POLE PLACEMENT CONTROLLER

Consider the following linear time invariant control law:

$$R(q^{-1})u(t) = S(q^{-1})(w(t) - y(t)), \tag{2}$$

where $w(t)$ is the reference signal.

The closed-loop system (1), (2) is:

$$y(t) = \frac{B(q^{-1})S(q^{-1})}{A(q^{-1})R(q^{-1}) + B(q^{-1})S(q^{-1})} w(t), \tag{3}$$

To carry out the design, the polynomial $B(\lambda)$ is factored as:

$$B(\lambda) = B^+(\lambda)B^-(\lambda),$$

where $B^+(\lambda)$ is a monic polynomial whose roots are outside the unit disc and well damped so that they can be cancelled by the controller. The degree of $B^-(\lambda)$ is denoted nb^-. We let $\mathbb{P}$ be the domain in which pole/zero cancellations are forbidden. In the sequel, for the sake of simplicity, we assume that $\mathbb{P}$ is a closed disc of radius $\alpha \geq 1$, $\bar{\mathbb{P}}$ will denote the complement of $\mathbb{P}$ in $\mathbb{C}$.

Pole placement consists in setting the denominator of the closed-loop transfer function (3) equal to a desired polynomial $A^*(\lambda)$. $A^*(\lambda)$ is called the desired closed-loop polynomial in the sequel. We assume that all its roots lie in $\bar{\mathbb{P}}$. Since $B^+(\lambda)$ is cancelled, $A^*(\lambda) = B^+(\lambda)A_1^*(\lambda)$, and $R(\lambda) = B^+(\lambda)R_1(\lambda)$. Hence the computation of the controller consists of solving the following Diophantine equation:

$$A(\lambda)R_1(\lambda) + B^-(\lambda)S(\lambda) = A_1^*(\lambda). \tag{4}$$

The coprimeness of $A(\lambda)$ and $B(\lambda)$ guarantees the existence of a solution to (4). There is actually an infinity of solutions to (4). This family of solutions can be parametrized as follows (Åström and Wittenmark, 1984; Vidyasagar, 1987a)

$$\begin{aligned} R_1(\lambda) &= R_1^0(\lambda) + X(\lambda)B^-(\lambda), \\ S(\lambda) &= S^0(\lambda) - X(\lambda)A(\lambda), \end{aligned} \tag{5}$$

where $(R_1^0(\lambda), S^0(\lambda))$ is a particular solution of (4), and $X(\lambda)$ is an arbitrary polynomial. Usually, the particular solution of (4) which is used in pole placement corresponds to a controller whose degree is: $\max(na - 1, nb - 1, na^* - \max(na, nb))$. This controller might however be unstable and/or nonminimum phase. In the sequel, we shall describe a method to design, for a specific class of plants, a stable controller that fulfils (4) starting from a solution of (4) that corresponds to an unstable controller. Actually, due to the design procedure, the poles of the stable controller will not only be located outside the unit disc, but they will lie outside $\mathbb{P}$.

From now on, a solution of (4) corresponding to a stable (unstable) controller will be called a stable (unstable) solution of (4).

4. POLE PLACEMENT WITH A STABLE CONTROLLER

4.1. *Stable controller design*

Theorem 1. We consider the class of plants (1), and the pole placement problem stated in Section 3. Its solution amounts to solving

equation (4) where $A_1^*(\lambda)$ has all its roots in $\bar{\mathbb{P}}$. There exists a stable solution to (4), corresponding to a controller with all its poles in $\bar{\mathbb{P}}$, if and only if the transfer function $P(\lambda)$ of the controlled system (1) has the parity interlacing property in the interval $[-\alpha, \alpha]$ (namely the number of real poles of $P(\lambda)$ between every pair of real zeros of $P(\lambda)$ in the interval $[-\alpha, \alpha]$ is even).

Proof.

IF: This part of the proof is constructive.

Let $(R_1^0(\lambda), S^0(\lambda))$ be a solution of (4) such that some roots of $R_1^0(\lambda)$ lie in $\mathbb{P}$. From (5), we know that all the polynomials $R_1(\lambda)$ that fulfil (4) satisfy:

$$\frac{R_1(\lambda) - R_1^0(\lambda)}{B^-(\lambda)} = X(\lambda), \quad \text{for some polynomial } X(\lambda). \tag{6}$$

It appears from (6) that a controller without pole in $\mathbb{P}$ can be obtained if and only if there exists a polynomial $R_1(\lambda)$, without root in $\mathbb{P}$, that interpolates $R_1^0(\lambda)$ at the roots of $B^-(\lambda)$. Now, notice that a polynomial without root in $\mathbb{P}$ is a particular unit of $R_{\bar{\mathbb{P}}}$ (the set of rational functions without pole in $\mathbb{P}$). Hence, our problem consists in finding a unit of $R_{\bar{\mathbb{P}}}$ with all its poles at infinity that interpolates $R_1^0(\lambda)$ at the roots of $B^-(\lambda)$. The interpolation problem can be stated precisely as follows:

(IP) We let $b_i, i = 1, \ldots, nd$, denote the distinct roots of $B^-(\lambda)$ arranged in such a way that the real roots are the first nr ones, and we let μ_i, $i = 1, \ldots, nd$ denote their multiplicities. We also introduce the following notations:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda^j} R_1^0(b_i) = r_{ij}, \quad j = 0, \ldots, \mu_i - 1; \quad i = 1, \ldots, nd, \tag{7}$$

where the zeroth order derivative of $R_1^0(\lambda)$ is taken as $R_1^0(\lambda)$ itself. The problem amounts to finding a polynomial unit $R_1(\lambda)$ of $R_{\bar{\mathbb{P}}}$ such that:

$$\frac{\mathrm{d}}{\mathrm{d}\lambda^j} R_1(b_i) = r_{ij}, \quad j = 0, \ldots, \mu_i - 1; \quad i = 1, \ldots, nd. \tag{8}$$

From Lemma 4.1 in Ghosh (1986), and from Vidyasagar (1987a), we know that the interpolation problem stated above has a solution if and only if the numbers r_{i0}, $i = 1, \ldots, nr$, are all of the same sign. Now, from (4) and (7) we derive:

$$r_{i0} = A_1^*(b_i)/A(b_i), \quad i = 1, \ldots, nr.$$

Since, by hypothesis, the roots of $A_1^*(\lambda)$ belong to $\bar{\mathbb{P}}$, the numbers $A_1^*(b_i)$, $= 1, \ldots, nr$, are all of the same sign. Thus for $r_{i0}, i = 1, \ldots, nr$, to be all of the same sign, $A(b_i), i = 1, \ldots, nr$, should all be of the same sign. This is assured if and only if the controlled system has the parity interlacing property with respect to the interval $[-\alpha, \alpha]$ where α is the radius of $\mathbb{P}$.

ONLY IF

Let $(R_1(\lambda), S(\lambda))$ be a solution of (4) such that the roots of $R_1(\lambda)$ lie inside $\bar{\mathbb{P}}$. Then, writing (4) for $\lambda = b_i$, $i = 1, \ldots, nr$ yields:

$$A(b_i)R_1(b_i) = A_1^*(b_i), \quad i = 1, \ldots, nr.$$

Since both $R_1(\lambda)$ and $A_1^*(\lambda)$ have all their roots in $\bar{\mathbb{P}}$, $R_1(b_i), i = 1, \ldots, nr$, are all of the same sign, and so are $A_1^*(b_i)$, $i = 1, \ldots, nr$. Hence $A(b_i), i = 1, \ldots, nr$, are all of the same sign and $P(\lambda)$ has the parity interlacing property in $[-\alpha, \alpha]$ (more details on the parity interlacing property can be found in Youla *et al.* (1974) and Vidyasagar (1987a)). □

The proof of Theorem 1 shows that the problem of designing a pole placement controller without pole in $\mathbb{P}$ amounts to the interpolation problem (IP). To solve (IP), we first notice that one can always recast the problem into another one, (IP1), which consists in building a unit of R_∞, the set of functions without pole inside $\mathbb{P}_1$ (i.e. inside the unit disc). Remembering that α is the radius of $\mathbb{P}$, we can state (IP1) as follows:

(IP1) Consider a set of numbers $b_i/\alpha \in \mathbb{P}_1$, $i = 1, \ldots, nd$. Define r_{ij} as in (7) and assume that r_{i0}, $i = 1, \ldots, nr$, all have the same sign. Then (IP1) consists of building a unit $\bar{R}_1(\lambda)$ of R_∞ with all its poles at infinity such that

$$\frac{\mathrm{d}}{\mathrm{d}\lambda^j} \bar{R}_1(b_i/\alpha) = r_{ij}, \quad j = 0, \ldots, \mu_i - 1; \quad i = 1, \ldots, nd. \tag{9}$$

It is straightforward to show that a solution to (IP) is:

$$R_1(\lambda) = \bar{R}_1(\lambda/\alpha). \tag{10}$$

Hence, without loss of generality, we can consider the problem (IP1) in the sequel. Moreover, in order to keep expressions simple, we let $\alpha = 1$.

4.2. *Interpolation with units in* R_∞

The main difference between (IP1) and the problem solved in Vidyasagar (1987a) lies in our requirement of a polynomial unit instead of a rational unit. We shall see that the procedure of Vidyasagar (1987a) can be modified easily to assure that the poles of the unit lie at infinity when the roots b_i are real. However, for a

complex b_i, the above mentioned method yields a denominator of the form $(\beta-\lambda)^a$ where $\beta>1$ is real and $a>0$ is an integer. We shall need the result described in the following theorem to cancel the pole at β.

Theorem 2. Let $b_i \in \mathbb{P}_1$, $i=1,\ldots,nr$ be real numbers and $b_i \in \mathbb{P}_1$, $i=nr+1,\ldots,j$ be complex numbers. Let β be real such that $|\beta|>1$. Define

$$p_j(\lambda)=\prod_{i=1}^{nr}(\lambda-b_i)\prod_{i=nr+1}^{j}(\lambda-b_i)(\lambda-\bar{b}_i). \quad (11)$$

Then the polynomial

$$E_j(\lambda)=1-(\lambda/\beta)^m p_j(\lambda)/p_j(\beta), \quad (12)$$

where $m>0$ is an integer such that

$$|\beta|^m > 2^{2j-nr}/|p_j(\beta)|, \quad (13)$$

has the following properties:
(1) $E_j(\beta)=0$,
(2) $E_j(b_i)=1$, $i=1,\ldots,j$,
(3) $E_j(\lambda)$ has no root in $\mathbb{P}_1$.

Proof. (1) and (2) obvious. (3) Notice that (13) can always be verified for sufficiently large m since $|\beta|>1$. (13) implies that:

$$\begin{aligned} &|\beta|^m > \|p_j(\lambda)\|_\infty/|p_j(\beta)| \quad \text{or} \\ &1/|\beta|^m < |p_j(\beta)|/\|p_j(\lambda)\|_\infty, \end{aligned} \quad (14)$$

since b_i, $i=1,\ldots,j$ lie inside the closed unit disc $\mathbb{P}_1$, and since $2j-nr$ is the number of roots of $p_j(\lambda)$. From (14),

$$|\lambda/\beta|^m < |p_j(\beta)|/\|p_j(\lambda)\|_\infty, \quad \forall \lambda \in \mathbb{P}_1,$$

and thus

$$|\lambda/\beta|^m \, \|p_j(\lambda)\|_\infty/|p_j(\beta)| < 1, \quad \forall \lambda \in \mathbb{P}_1.$$

This yields

$$|(\lambda/\beta)^m p_j(\lambda)/p_j(\beta)| < 1, \quad \forall \lambda \in \mathbb{P}_1,$$

which implies that the roots of $E_j(\lambda)$ lie outside $\mathbb{P}_1$. □

We are now ready to describe the modifications to be made in Vidyasagar (1987a) to obtain a procedure to solve (IP1). Due to a lack of space, we only consider the case $\mu_i=1$, $i=1,\ldots,nd$, which corresponds to a situation where all the roots of $B^-(\lambda)$ are distinct. As in Vidyasagar (1987a), we arrange the complex roots so that b_i, $i=nr+1,\ldots,(nr+nb^-)/2$ have positive imaginary parts.

First suppose that we have found a unit $u(\lambda)$ with all its poles at infinity such that:

$$u(b_i)=R_1^0(b_i)=r_{i0}, \quad i=1,\ldots,j-1<nr. \quad (15)$$

We show how to construct a unit v that satisfies (15) with u replaced by v, and in addition, satisfies $v(b_j)=R_1^0(b_j)=r_{j0}$. If $j=1$, so that condition (15) is vacuous, simply choose $v=r_{10}$. Otherwise, proceed as follows: define

$$f(\lambda)=\prod_{i=1}^{j-1}(\lambda-b_i). \quad (16)$$

Then $f(\lambda)\in R_\infty$; moreover $1+cf$ is a unit whenever $|c|<1/\|f\|_\infty$. For such a c the function $(1+cf)^d$ is also a unit with all its poles at infinity for all positive integers d. c and d can be selected so that $|c|<1/\|f\|_\infty$ and so that

$$v=(1+cf)^d u, \quad (17)$$

satisfies

$$v(b_j)=r_{j0}. \quad (18)$$

Following Vidyasagar (1987a), c is given by:

$$c=([r_{j0}/u(b_j)]^{1/d}-1)/f(b_j). \quad (19)$$

Suppose now that we have found a unit with poles at infinity u such that

$$\begin{aligned} &u(b_i)=R_1^0(b_i)=r_{i0}, \\ &i=1,\ldots,j-1; \quad nr<j<(nb^-+nr)/2+1. \end{aligned} \quad (20)$$

We show how to construct a unit η which satisfies (20) with u replaced by η, and in addition satisfies

$$\eta(b_j)=R_1^0(b_j)=r_{j0}. \quad (21)$$

By straightforward extension of Vidyasagar (1987a) we obtain the following form for $\eta(\lambda)$

$$\eta(\lambda)=(1+bf(\lambda))^a u(\lambda), \quad (22)$$

where

$$f(\lambda)=\{(\beta\lambda-1)/(\beta-\lambda)\}p_{j-1}(\lambda), \quad \beta>1, \quad (23)$$

with $p_{j-1}(\lambda)$ defined by (11) where j is substituted by $j-1$.

The real numbers b and β and the integer a can be determined in such a way that (21) is fulfilled. Hence, $\eta(\lambda)$ is a unit which satisfies the required interpolation constraints. However, due to the form of $f(\lambda)$ in (23), it has a pole at $\lambda=\beta>1$ with multiplicity a. We must cancel this pole in order to obtain a polynomial unit. To this end, notice that

$$v(\lambda)=\eta(\lambda)(E_j(\lambda))^a, \quad (24)$$

where $E_j(\lambda)$ is defined by (11), (12) and (13), is a unit which fulfils (20), (21), with respectively, u and η replaced by v. Furthermore, $v(\lambda)$ has all its poles at infinity. Indeed, by Theorem 2.3, $v(\lambda)$ is a unit, and by Theorem 2.2, $v(b_i)=\eta(b_i)$, $i=1,\ldots,j$; moreover, from Theorem 2.1 $(E_j(\lambda))^a$ has a root at β with

multiplicity a, which cancels the corresponding pole in η, so that the above claims are proved.

Applying iteratively the procedure described above for $j = 1, \ldots, nb^-$ yields a solution to (IP1).

5. SIMULATION RESULTS

5.1. *Numerical example*

Consider a plant with transfer function $P(s) = (s - 10)/[s(s - 1)]$. We would like to design a discrete time controller so that the closed-loop bandwidth is 4 rad sec^{-1}. We use a fast sampling policy and we choose the sampling period $T_s = 0.02$ s. The transfer function of the sampled system is

$$P(\lambda) = [\lambda(0.0182 - 0.0222\lambda)]/(1 - 2.0202\lambda + 1.0202\lambda^2) = B(\lambda)/A(\lambda). \quad (25)$$

It has one pole and two zeros inside the unit circle, respectively at 0.9802, 0.8182 and 0.

In order to achieve the desired closed-loop bandwidth, we choose the following desired closed-loop polynomial:

$$A^*(\lambda) = 1 - 1.8462\lambda + 0.8521\lambda^2. \quad (26)$$

It has two roots at $\lambda = \exp(4 * 0.02) = 1.0833$. No zero of $B(\lambda)$ can be cancelled. $\mathbb{P}$ is chosen as the closed unit disc. Thus $\alpha = 1$.

The particular solution of the Diophantine equation (4) corresponding to a controller of order $\max(na - 1, nb - 1, na^* - \max(na, nb))$ i.e. of order 1 here, yields the following unstable controller:

$$S^0(\lambda)/R^0(\lambda) = (-57.1774 + 55.7144\lambda)/(1 + 1.2139\lambda). \quad (27)$$

We now apply the procedure of Section 4.2 to design a stable controller achieving the desired closed-loop polynomial (26). The roots of $B(\lambda)$ are $b_1 = 0$ and $b_2 = 0.8182$. We deduce $r_{10} = 1$, $r_{20} = 1.9933$. r_{10} and r_{20} are of the same sign; thus the problem can be solved. A unit that interpolates $R^0(\lambda)$ at 0 is $u(\lambda) = 1$. From (16), $f(\lambda) = \lambda$. (19) with $j = 2$ yields $c = 0.5033$ for $d = 2$. Obviously $|c| < 1/\|f\|_\infty = 1$. Thus the unit that solves our problem, which is the denominator of the stable controller we are looking for, is:

$$R(\lambda) = (1 + 0.5033\lambda)^2. \quad (28a)$$

From (6), we deduce $X(\lambda) = -11.3966$, and from (5).

$$S(\lambda) = -45.7809 + 32.6910\lambda + 11.6268\lambda^2. \quad (28b)$$

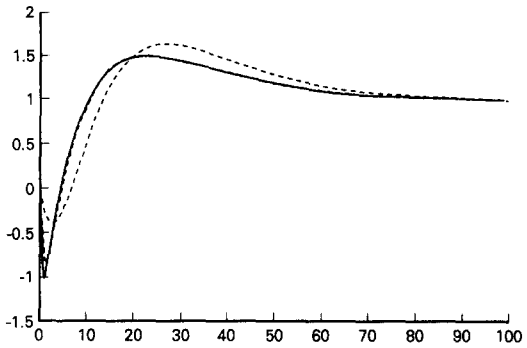


FIG. 2. Step responses of the closed-loop system obtained in Example 5.1. —— with controller (27), --- with controller (28), -·-·- with controller (29). Time axis graduated in sampling periods.

The closed-loop system transfer functions $T(\lambda)$ and $T_0(\lambda)$, corresponding to the controllers (28) and (27), respectively, can be computed via (3). Due to the fact that the stable controller (the SCI) has higher degree than the unstable one, $T(\lambda)$ has two poles at infinity while $T_0(\lambda)$ only has one pole at infinity.

Figure 2 gives the step responses of $T_0(\lambda)$ and $T(\lambda)$. The use of a SCI yields a 20% decrease of the undershoot without modifying significantly the other characteristics of the step response. The gain margins for both systems are almost the same. With the unstable controller, the loop gain can be multiplied by any $k \in [0.23, 1.78]$ without affecting closed-loop stability. With the stable controller $k \in [0.24, 1.78]$. The better measurement noise rejection obtained with the SCI appears clearly in Fig. 3, where the frequency responses of T_0 and T are represented (Doyle and Stein, 1981). The SCI can cope better than the unstable controller with high-frequency (larger than 20 rad sec^{-1}) measurement noise.

Notice that another solution to avoid using an unstable controller is to modify the desired closed-loop polynomial $A^*(\lambda)$ (SCACP design).

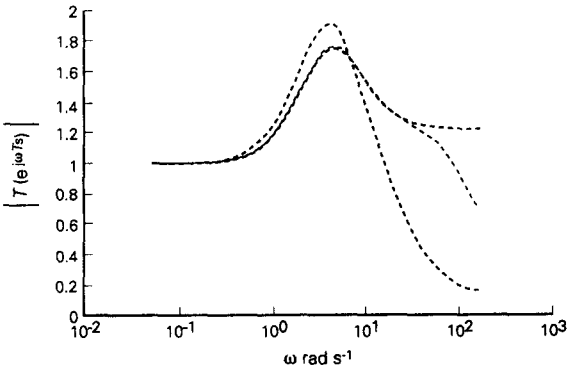


FIG. 3. Frequency responses of the closed-loop systems considered in Fig. 2. ($|T(e^{j\omega})|$ vs ω in rad sec^{-1}). The correspondence between the line-types and the systems are the same as in Fig. 2.

If, as suggested by the guidelines of Middleton and Goodwin (1990) we choose $a1 = a2 = \exp(4 * 0.02)$ and $a3 = \exp(12 * 0.02)$ as roots for $A^*(\lambda)$, then the classical solution of the Diophantine equation yields the stable controller:

$$S_c(\lambda)/R_c(\lambda) = (-14.2188 + 13.9066\lambda)/(1 - 0.3540\lambda). \quad (29)$$

The corresponding closed-loop step response has a smaller undershoot, a larger overshoot and a larger settling time compared with the previous designs (see Fig. 2). The loop gain can be multiplied by a gain $k \in [0.27, 1.97]$ without losing stability. From Fig. 3, it appears that this controller can cope better than the previous designs with measurement noise above 10 Hz, while any measurement noise within a frequency range inside the desired bandwidth will affect the tracking error more seriously.

5.2. *Discussion*

Other simulations have shown that a SCI does not always yield a faster step response transient than a SCACP built according to the guidelines of Middleton and Goodwin (1990) for a given desired closed-loop bandwidth. Slower transients can occur when the interpolation procedure yields a large increase of the controller degree, compared with the degree of the unstable controller obtained by classical pole placement design. On the other hand the gain margins obtained with an SCI might be better than those obtained with an SCACP. Hence, general conclusions cannot be deduced as to what method is most suitable for given performance improvement requirements. However, we believe that the design of SCI offers a useful complement to the method consisting in an adequate modification of the desired closed-loop characteristic polynomial, because it is systematic while the other approach is based on heuristic rules which might still require some refinement.

The main drawback of SCI design, namely the risk of obtaining high order controllers, might be partly alleviated by trying to adapt to discrete-time the method to build low order interpolating units in H_∞ of Dorato *et al.* (1989). However, this problem is not straightforward because we require a polynomial unit in R_∞ and Dorato's procedure does not allow one to choose the poles of the unit. Theorem 2 could be used to cancel the unwanted finite poles, but this would destroy at least part of the benefit of this approach.

The simulation results clearly illustrate the importance of using stable controllers. For the plant models that we considered, such controllers always improve at least one of the following properties of the closed-loop system, with respect to their unstable counterpart: transient performances (step response), robustness, measurement noise rejection.

6. SIMULTANEOUS PARTIAL POLE PLACEMENT

6.1. *Problem statement*

(SPPP) Let $P_0(\lambda) = B_0(\lambda)/A_0(\lambda)$ and $P_1(\lambda) = B_1(\lambda)/A_1(\lambda)$ be the transfer functions of two distinct plants of the class (1). (SPPP) consists in designing a controller that achieves pole placement, such as described in Section 3, for the plant $P_1(\lambda)$ while assuring that the same controller yields a closed-loop system with all its poles in $\bar{\mathbb{P}}$ for the plant $P_0(\lambda)$. $A^*(\lambda)$ denotes the desired closed-loop polynomial for plant $P_1(\lambda)$. All its roots lie in $\bar{\mathbb{P}}$ and $A^*(\lambda) = B_1^+(\lambda)A_1^*(\lambda)$ where $B_1(\lambda) = B_1^+(\lambda)B_1^-(\lambda)$, as in Section 3.

If the roots of $A^*(\lambda)$ where located outside $\bar{\mathbb{P}}$, the problem could be formulated as a particular case of the simultaneous partial pole placement of Ghosh (1986). However, since this condition is not verified, (SPPP) cannot be solved in this way. Notice also that (SPPP) cannot be reformulated in the framework of Vidyasagar's approach. Indeed, Vidyasagar only requires that the closed-loop poles lie in an "admissible region" (an open set) and not at some precise location (namely the roots of $A^*(\lambda)$).

6.2. *Solution*

Before giving the solution of (SPPP), we review the following fundamental result.

Theorem 3 (Vidyasagar and Viswanadham, 1982; Vidyasagar, 1987a). Consider two plants $P_0(\lambda)$ and $P_1(\lambda)$ as in (SPPP) above. $P_0(\lambda)$ and $P_1(\lambda)$ can be simultaneously stabilized if and only if the plant $P(\lambda)$ defined below is strongly stabilizable:

$$P(\lambda) = B(\lambda)/A(\lambda) = [A_0(\lambda)B_1(\lambda) - B_0(\lambda)A_1(\lambda)]/[Y_0(\lambda)A_1(\lambda) + X_0(\lambda)B_1(\lambda)], \quad (30)$$

where $X_0(\lambda)$ and $Y_0(\lambda)$ are polynomials in λ that fulfil

$$Y_0(\lambda)A_0(\lambda) + X_0(\lambda)B_0(\lambda) = I. \quad (31)$$

Moreover, if $C(\lambda)$ is a stable controller that stabilizes $P(\lambda)$, then a controller that achieves simultaneous stabilization of $P_0(\lambda)$ and $P_1(\lambda)$ is given by:

$$C_{sim}(\lambda) = [X_0(\lambda) + C(\lambda)A_0(\lambda)]/[Y_0(\lambda) - C(\lambda)B_0(\lambda)]. \quad (32)$$

Theorem 4. Consider two plants $P_0(\lambda)$ and $P_1(\lambda)$ as in (SPPP) above, and build $P(\lambda)$ as in (30). Then (SPPP) has a solution if and only if $P(\lambda)$ fulfils the parity interlacing property in the interval $[-\alpha, \alpha]$ (where α is the radius of $\mathbb{P}$).

Proof. From Theorem 1, we know that, for $P(\lambda)$, there exists a controller without pole in $\mathbb{P}$, $C_0(\lambda) = S_0(\lambda)/R_0(\lambda)$, that yields the desired closed-loop polynomial $A^*(\lambda)$ if and only if $P(\lambda)$ fulfils the parity interlacing property in $[-\alpha, \alpha]$. We are going to show that, if $C(\lambda)$ is replaced by $C_0(\lambda)$ in (32), the resulting controller, $C_{\text{sim}}(\lambda)$ gives a solution to (SPPP). To this end, first notice that, since $C_0(\lambda)$ is stable, $C_{\text{sim}}(\lambda)$ achieves simultaneous stabilization of $P_0(\lambda)$ and $P_1(\lambda)$ (see Theorem 3). Hence, from Vidyasagar (1987a) $C_{\text{sim}}(\lambda)$ can also be expressed in terms of the plant $P_1(\lambda)$ as follows:

$$C_{\text{sim}}(\lambda) = [Y_1(\lambda) + C_1(\lambda)A_1(\lambda)]/[Y_1(\lambda) - C_1(\lambda)B_1(\lambda)], \quad (33)$$

where

$$Y_1(\lambda)A_1(\lambda) + X_1(\lambda)B_1(\lambda) = I,$$

and

$$C_1(\lambda) = U^{-1}(\lambda)[V(\lambda) + C_0(\lambda)W(\lambda)], \quad (34)$$

with

$$U(\lambda) = A(\lambda) + C_0(\lambda)B(\lambda), \quad (35)$$

$$\begin{bmatrix} V(\lambda) \\ W(\lambda) \end{bmatrix} = \begin{bmatrix} Y_0(\lambda) & X_0(\lambda) \\ -B_0(\lambda) & A_0(\lambda) \end{bmatrix} \begin{bmatrix} -X_1(\lambda) \\ Y_1(\lambda) \end{bmatrix}. \quad (36)$$

By construction of $C_0(\lambda)$, (35) can be rewritten:

$$U(\lambda) = A^*(\lambda)/R_0(\lambda). \quad (37)$$

Then (34) and (37) yield:

$$C_1(\lambda) = [V(\lambda)R_0(\lambda) + W(\lambda)S_0(\lambda)]/A^*(\lambda). \quad (38)$$

We can now compute the closed-loop systems obtained by using $C_{\text{sim}}(\lambda)$ to control $P_0(\lambda)$ and $P_1(\lambda)$. We denote them $T_0(\lambda)$ and $T_1(\lambda)$, respectively. Using (32) with $C(\lambda)$ replaced by $C_0(\lambda)$, we obtain:

$$\begin{aligned} T_0(\lambda) &= B_0(\lambda)[X_0(\lambda) + C_0(\lambda)A_0(\lambda)] \\ &= B_0(\lambda)[X_0(\lambda)R_0(\lambda) + A_0(\lambda)S_0(\lambda)]/R_0(\lambda). \end{aligned} \quad (39)$$

With the controller (33) and (38), we compute:

$$\begin{aligned} T_1(\lambda) &= B_1(\lambda)[X_1(\lambda) + C_1(\lambda)A_1(\lambda)] \\ &= B_1(\lambda)[X_1(\lambda)A^*(\lambda) + (V(\lambda)R_0(\lambda) \\ &\quad + W(\lambda)S_0(\lambda))A_1(\lambda)]/A^*(\lambda). \end{aligned} \quad (40)$$

Since the roots of $R_0(\lambda)$ are located in $\bar{\mathbb{P}}$ by construction, (39) and (40) conclude the proof. Notice that state space expressions for (39) and (40) were previously derived in Vidyasagar (1988) but not for purpose of simultaneous partial pole placement.

An advantage of considering the problem (SPPP) instead of the approach of Ghosh or Vidyasagar is that, in some circumstances the (SPPP) might be solved without any interpolation procedure. For instance, when $\bar{\mathbb{P}}$ is simply the complement of the closed unit disc, the guidelines of Middleton and Goodwin (1990) could be used to state a (SPPP), for a given couple of plants, that can be solved by computing the classical solution of a Diophantine equation. Indeed, those guidelines give conditions on the desired closed-loop polynomial to assure that $C_0(\lambda)$, as obtained via the classical pole placement procedure, be stable for a given $P(\lambda)$. This approach is computationally very simple, compared with interpolation procedures.

7. CONCLUSION

For plants satisfying a specific parity interlacing property, we have presented a procedure to achieve pole placement with a stable controller, when the classical pole placement procedure yields an unstable controller. Simulation results have shown that the stability of the controller leads improved closed-loop performances. Ways to modify the initial desired closed-loop polynomial so as to obtain a stable controller with the classical pole placement method have also been discussed in the numerical example.

Of course, when we speak about stable controllers, we do not exclude controllers with single or multiple integrating actions. Such controllers are built by introducing the required integrating actions in the plant model and performing the procedure of Section 4 on the augmented plant.

The design of SCI can be modified in an obvious way to derive a solution of the Diophantine equation (4) corresponding to a minimum phase controller, from a solution that corresponds to a non-minimum phase controller. The design of a stable and minimum phase controller that achieves pole placement is still an open problem. A first step toward its solution is described in Wei (1990) where stabilization with a stable continuous-time controller having no real non-minimum phase zeros is considered. Wei gives necessary and sufficient solvability conditions for this problem, and he describes a design procedure which guarantees that the controller has the desired properties. It is however proved in Blondel *et al.* (1991) that

these conditions are, in general, not sufficient for stabilization by a stable and minimum phase controller.

Finally, we have shown that the method to obtain an SCI can be used to solve a particular simultaneous partial pole assignment problem (SPPP). An advantage of (SPPP), compared with Ghosh (1986) or Vidyasagar's (1987b) approach, is that it can, in some circumstances, be solved without any interpolation procedure, which significantly simplifies the computations.

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